

1. Project Objective

FinPulse is a full-stack stock market intelligence platform developed to monitor 20 NSE-listed companies, store historical market data ,expose REST APIs using FastAPI and visualize market information through an interactive React dashboard.

2. System Architecture

The application follows a three-tier architecture:

- Frontend: React + Vite for the user interface.
- Backend: FastAPI providing REST APIs and business logic.
- Database: SQLite for storing company fundamentals and historical OHLCV data.
- External Data Source: Yahoo Finance (yFinance).

3. Features Implemented

Requirement	Implementation
Track 20 companies	Tracks 20 NSE-listed companies
Database	SQLite stores fundamentals and historical prices
REST APIs	7 API endpoints for retrieval and updates
Historical Charts	Interactive line and candlestick charts
Fundamentals	Price, Market Cap, P/E Ratio, EPS
Comparison	Compare multiple companies
Dashboard	Interactive React dashboard with screener and market summary

4. Database Design

Table: stocks

• ticker • company_name • price • market_cap • pe_ratio • eps • return_percent

Table: price_history

• ticker • date • open • high • low • close • volume

5. REST API Endpoints

GET /stocks

GET /stocks/{ticker}

GET /stocks/{ticker}/history

GET /market-summary

GET /compare?tickers=A,B,C

POST /update/{ticker}

POST /update-all

6. Challenges

The primary challenges included integrating React with FastAPI, handling incomplete responses from Yahoo Finance, designing an efficient schema for historical data, and rendering interactive financial charts while keeping the application responsive.

7. Future Improvements

Potential future enhancements include portfolio tracking, sector-wise analytics, real-time streaming market updates, user authentication, and downloadable reports.

8. Conclusion

FinPulse successfully satisfies the assignment objectives by integrating data acquisition, persistent storage, REST APIs, and interactive visualization into a complete full-stack application. The project demonstrates practical experience with modern web development technologies and financial data processing.

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